

Cem Ozcelik

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PROFESSIONAL SUMMARY

Senior Data Scientist with an Industrial Engineering background and nearly 5 years of progressive experience building advanced predictive analytics models on large datasets. Deep stack across **Python, deep learning (CNN, RNN, LSTM in TensorFlow/Keras and PyTorch), advanced SQL, and distributed ML on PySpark/Databricks**. Built a production-ready microservices search engine on 800K+ product descriptions with SentenceTransformers, custom Word2Vec, Item2Vec FBT recommendations, and an LSTM next-token autocomplete -- the same architectural pattern that powers marketplace search at scale. Computer vision delivered too: pneumonia detection with VGG16 fine-tuning (97% Recall, 93% Test Accuracy) and Grad-CAM explainability. **Daily AI-augmented development workflow** using Claude Code / Cursor for scaffolding, testing, and refactoring -- with human judgment retained for design and trade-offs. Comfortable communicating data findings clearly across cross-functional squads in Turkish and English. Demo:

github.com/ccemozclk/amazon-ai-search-engine | github.com/ccemozclk/Spark-Recommendation-Engine

CORE COMPETENCIES

Skills: Deep Learning (CNN / RNN / LSTM), Search & Recommendation, Computer Vision, Statistical Modelling, Distributed ML (PySpark / Databricks), Model Monitoring, A/B Testing, Cross-Functional Squad Communication (TR/EN)

Tools: Python (Advanced), SQL (Advanced), TensorFlow, Keras, PyTorch, PySpark, Databricks, FastAPI, Docker, Qdrant (Vector DB), Redis, Claude Code, Cursor, Git, AWS (EC2/S3)

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 -- Present

Senior Database Analyst | Maltepe, Istanbul

- Took **full ownership of an end-to-end ML platform**, scaling an internal Django + Docker application into a hub serving versioned REST APIs for production analytics and ML models.
- Mined and analyzed internal databases to **support product, optimize operational strategy, and improve business outcomes**, including SKU-level Association Rule Mining for cross-sell strategy.
- Directed predictive supply chain analytics in Python and advanced SQL, sustaining an **8-hour weekly reduction in operational overtime** through ML-driven forecasting.
- Collaborated within cross-functional squads (engineering, operations, business) translating model outputs into clear insights for technical and non-technical stakeholders.
- **Tech:** Python, SQL, Django, REST APIs, ML Pipelines, Docker, PostgreSQL, Association Rule Mining

Anadolu Pet Akvaryum & Effeffe

Sept 2024 -- Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Architected and deployed a full-stack Django web application from scratch (SOLID principles), establishing the foundation that later scaled into the production ML platform.
- Engineered predictive supply chain models in Python and PL/SQL that **reduced stockouts and improved inventory turnover**.
- Conducted Route-Level Profitability Analysis with advanced SQL, delivering a **3% improvement in overall logistics efficiency**.
- **Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive

Jan 2022 -- Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Developed a **Predictive Maintenance & Anomaly Detection framework** using unsupervised clustering on machine-sensor time-series -- the same statistical primitives that drive listing-anomaly and fraud-listing detection on marketplace platforms.
- Delivered a **15% weekly increase in critical machine availability** and a **15% improvement in OEE**, validated by operations leadership on the factory floor.
- Engineered robust ETL pipelines in MSSQL and Python for high-volume sensor data, replacing manual reporting with reliable analytical infrastructure.
- Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series analysis, anomaly detection

Birinci Automotive

June 2021 -- Jan 2022

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical methods to workforce datasets, uncovering hidden patterns to optimize HR policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

E-Commerce AI Search Engine Semantic Search & Marketplace Recommendation

Production-ready **microservices architecture** with dual-engine A/B testing: SentenceTransformers (all-MiniLM-L6-v2) vs custom-trained Word2Vec on **800K+ product descriptions**. Custom **LSTM next-token prediction** in PyTorch for intelligent search auto-complete. **Item2Vec collaborative filtering** for "Frequently Bought Together" / "Similar listings buyers also viewed" recommendations across hundreds of thousands of user sessions. Sub-millisecond cached responses via Redis; PySpark distributed ETL for massive JSONL ingestion. Same architectural pattern transfers 1:1 to marketplace search ranking and listing recommendation.

PyTorch, FastAPI, Streamlit, Qdrant (Vector DB), Redis, PySpark, Docker, SentenceTransformers | github.com/ccemozclk/amazon-ai-search-engine

Scalable Recommendation Engine PySpark & Customer Segmentation

End-to-end personalized recommendation system on **Apache Spark / Databricks** (Online Retail II). **RFM analysis + K-Means** for behavioural customer segmentation into 5 strategic tiers (Champions, At-Risk, etc.) -- direct analog to dealer/buyer tiering and lifecycle modelling on a marketplace. **ALS collaborative filtering** with implicit feedback (purchase quantity proxy) and explicit cold-start handling. Hybrid content + collaborative approach validated by theme-affinity tests.

PySpark, Apache Spark, Databricks, K-Means, ALS, Streamlit | github.com/ccemozclk/Spark-Recommendation-Engine

AI-Powered Medical Diagnostics Computer Vision & Explainable AI

97% Recall on pneumonia detection from chest X-rays -- explicitly optimized for asymmetric error cost (False Negative >> False Positive). **93% Test Accuracy** with Fine-Tuned VGG16 (vs 76% baseline CNN, 81% ResNet50). Implemented **Grad-CAM** for visual explainability and used it to identify and correct a dataset bias invisible without heatmap analysis. Class-weight balancing + surgical fine-tuning of `block5` at LR=1e-5. Same CV techniques transfer directly to car-listing image quality, body-damage detection, and duplicate-image fraud detection.

Python, TensorFlow-Keras, VGG16, ResNet50, OpenCV, Grad-CAM | github.com/ccemozclk/Deep-Learning-Project-CNN-VGG16-ResNet50

EDUCATION

B.Sc. Industrial Engineering — Kirikkale University

Sept 2016 -- Feb 2021

Quantitative degree (Mathematics, Engineering, Statistics core) with rigorous foundation in statistical analysis, operations research, and optimization. Designed MILP models with IBM ILOG CPLEX; applied AHP and TOPSIS to multi-criteria decision problems -- direct prerequisite for cost-function-aware modelling and trade-off design.

CERTIFICATIONS

Associate Data Scientist Career Track (incl. PySpark, Recommendation Engines) — [DataCamp](#)

Jan -- May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — [Miuul](#)

Mar -- Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash

Deep Learning: CNN, RNN, LSTM (Bidirectional + custom architectures), TensorFlow, Keras, PyTorch, Transfer Learning, Fine-Tuning

Computer Vision: VGG16, ResNet50, OpenCV, Grad-CAM (Explainable AI), Image Classification

Search & Rec: SentenceTransformers, Word2Vec, Item2Vec, ALS, K-Means, RFM, Qdrant (Vector DB)

Classical ML: XGBoost, LightGBM, scikit-learn, h2o.ai, A/B Testing, Statistical Modelling, Feature Engineering

Big Data: PySpark, Apache Spark, Databricks, Distributed Computing

MLOps & AI-Augmented Dev: Docker, FastAPI, Django, Redis, REST APIs, Git, AWS (EC2/S3), Claude Code, Cursor

Databases: PostgreSQL, MSSQL, Oracle, ETL Pipelines, Pandas, NumPy **Languages:** Turkish (Native), English (Fluent)